LEC - 50

Earners AI and ML by Aditya Jain sir -Numpy

SUMMARY NOTES

```
a = [3, 5.6, "AJ", True]
type(a)
```

```
list
```

```
print(a)
```

```
[3, 5.6, 'AJ', True]
```

```
import numpy as np
b = np.array(a)
print(b)
```

```
['3' '5.6' 'AJ' 'True']
```

```
a = [4,6,2,9]
b = np.array(a)
type(a)
```

```
list
```

```
type(b)
```

```
numpy.ndarray
```

```
a+2
```

```
---------------------------------------------------------------------------
TypeError                                 Traceback (most recent call last)
/tmp/ipykernel_2301/2232802120.py in <cell line: 0>()
----> 1 a+2

TypeError: can only concatenate list (not "int") to list
```

Next steps: Explain error

```
b+2
```

```
array([ 6,  8,  4, 11])
```

```
b*2
```

```
array([ 8, 12,  4, 18])
```

```
b-2
```

```
array([2, 4, 0, 7])
```

```
b**2
```

```
array([16, 36,  4, 81])
```

```
a = np.array([4,3,1])
b = np.array([5,4,2])

a+b
```

```
array([9, 7, 3])
```

```
a-b
```

```
array([-1, -1, -1])
```

```
a*b
```

```
array([20, 12,  2])
```

```
#Given a list of prices, update them with gst included prices.

prices = [50, 100, 40, 90, 80]

gst_p = []

for p in prices:
  gst_p.append(p+0.18*p)

print(gst_p)
```

```
[59.0, 118.0, 47.2, 106.2, 94.4]
```

```
np_prices = np.array(prices)

gst_np = np_prices*1.18

print(gst_np)
```

```
[ 59.  118.   47.2 106.2  94.4]
```

```
v = np.arange(1,10)
print(v)
```

```
[1 2 3 4 5 6 7 8 9]
```

```
np.linspace(0,10,5)
```

```
array([ 0. ,  2.5,  5. ,  7.5, 10. ])
```

```
0 ---- 10
0 --> 2.5 --> 5 --> 7.5 --> 10
```

```
# p + 18%p = p+0.18*p = p(1+0.18) = 1.18*p
```

```
num = (4,6,3,1)
a = np.array(num)
print(a)
```

```
[4 6 3 1]
```

```
type(a)
```

```
numpy.ndarray
```

```
type(num)
```

```
tuple
```

```
a = np.array([5,4,5.5,9])
```

```
print(a)
```

```
[5.  4.  5.5 9. ]
```

```
names = np.array(["AJ", "Riya", "Raj", "Rahul"])
```

```
print(names)
```

```
['AJ' 'Riya' 'Raj' 'Rahul']
```

```
type(names)
```

```
numpy.ndarray
```

```
a = np.array([True, False, False])
print(a)
```

```
[ True False False]
```

```
type(a)
```

```
numpy.ndarray
```

```
marks = np.array([
    [70, 80, 90],
    [60, 75, 85]
])
```

```
marks[0,1]
```

```
np.int64(80)
```

```
marks[0][1]
```

```
np.int64(80)
```

```
np.zeros(4)
```

```
array([0., 0., 0., 0.])
```

```
np.ones(5)
```

```
array([1., 1., 1., 1., 1.])
```

```
np.ones(5)*5
```

```
array([5., 5., 5., 5., 5.])
```

```
np.full(6,2)
```

```
array([2, 2, 2, 2, 2, 2])
```

```
np.full((3,4), 2)
```

```
array([[2, 2, 2, 2],
       [2, 2, 2, 2],
       [2, 2, 2, 2]])
```

```
# 0 - 10 all even numbers:
# linspace, arange


a = np.arange(0,11,2)
print(a)
```

```
[ 0  2  4  6  8 10]
```

```
b = np.linspace(0,10,6)
print(b)
```

```
[ 0.  2.  4.  6.  8. 10.]
```

```
print(b.dtype)
```

```
float64
```

```
a.dtype
```

```
dtype('int64')
```

```
a = np.array([True,False])
a.dtype
```

```
dtype('bool')
```

```
b  = np.array([4,6,3.5])
b.dtype
```

```
dtype('float64')
```

```
a = np.array([10,20,30], dtype = float)
print(a)
```

```
[10. 20. 30.]
```

```
a.dtype
```

```
dtype('float64')
```

```
b = np.array([10.5,4.7,7], dtype = int)
print(b)
```

```
[10  4  7]
```

```
c = np.array([2,4,0,7,0], dtype = bool)
print(c)
```

```
[ True  True False  True False]
```

```
# 0 -> False
# non-zero ->True
```

```
import numpy as np

marks = np.array([60, 70, 80, 90, 100])
bonus = marks + 5
final_marks = bonus * 1
print(marks)
print(bonus)
print(final_marks)
```

```
[ 60  70  80  90 100]
[ 65  75  85  95 105]
[ 65  75  85  95 105]
```

```
import numpy as np

numbers = [10, 20, 30]
list_result = numbers * 2

arr = np.array(numbers)
array_result = arr * 2

print(list_result)
print(array_result)
```

```
[10, 20, 30, 10, 20, 30]
[20 40 60]
```

```
import numpy as np

prices = np.array([100, 200, 300, 400, 500])
discount = prices * 0.10
final_price = prices - discount

print(discount)
print(final_price)
```

```
[10. 20. 30. 40. 50.]
[ 90. 180. 270. 360. 450.]
```

```
numbers = np.arange(2, 21, 3)
print(numbers)
```

```
[ 2  5  8 11 14 17 20]
```

```
import numpy as np

numbers = np.arange(2, 21, 3)
print(numbers)
```

```
print(numbers[0])
print(numbers[-1])
print(len(numbers))
print(numbers * 2)
```

```
[ 2  5  8 11 14 17 20]
2
20
7
[ 4 10 16 22 28 34 40]
```

```
Start coding or generate with AI.
```

```
values = np.linspace(0, 20, 5)
print(values)
```

```
[ 0.  5. 10. 15. 20.]
```

```
import numpy as np

values = np.linspace(0, 20, 5)
print(values)
doubled = values * 2
print(doubled)
print(values + 10)
print(values / 2)
```

```
[ 0.  5. 10. 15. 20.]
[ 0. 10. 20. 30. 40.]
[10. 15. 20. 25. 30.]
[ 0.   2.5  5.   7.5 10. ]
```

```
import numpy as np

attendance = np.zeros(5)
print(attendance)

attendance = attendance + 5
print(attendance)

attendance = attendance + 3
print(attendance)

attendance = attendance * 2
print(attendance)
```

```
[0. 0. 0. 0. 0.]
[5. 5. 5. 5. 5.]
[8. 8. 8. 8. 8.]
[16. 16. 16. 16. 16.]
```

```
import numpy as np

salary = np.full(5, 30000)
print(salary)

increment = salary * 0.10
print(increment)

new_salary = salary + increment
print(new_salary)
print(new_salary * 2)
```

```
[30000 30000 30000 30000 30000]
[3000. 3000. 3000. 3000. 3000.]
[33000. 33000. 33000. 33000. 33000.]
[66000. 66000. 66000. 66000. 66000.]
```

```
import numpy as np

arr = np.array([10, 20, 30.5, 40, 50])
print(arr)
print(arr.dtype)

result = arr * 2
```

```
print(result)
print(result.dtype)
```

```
[10.  20.  30.5 40.  50. ]
float64
[ 20.  40.  61.  80. 100.]
float64
```

```
import numpy as np

marks = np.array([
    [70, 80, 90],
    [60, 75, 85]
])

print(marks)
print(marks + 5)
print(marks * 2)
print(marks - 10)
```

```
[[70 80 90]
 [60 75 85]]
[[75 85 95]
 [65 80 90]]
[[140 160 180]
 [120 150 170]]
[[60 70 80]
 [50 65 75]]
```

```
import numpy as np

a = np.array([10, 20, 30, 40])
b = np.array([1, 2, 3, 4])

addition = a + b
subtraction = a - b
multiplication = a * b

print(addition)
print(subtraction)
print(multiplication)
```

```
[11 22 33 44]
[ 9 18 27 36]
[ 10  40  90 160]
```

```
a = np.array([10, 20, 30, 40])
b = np.array([1, 2, 3])

addition = a + b
subtraction = a - b
multiplication = a * b

print(addition)
```

```
---------------------------------------------------------------------------
ValueError                                Traceback (most recent call last)
/tmp/ipykernel_2301/357092748.py in <cell line: 0>()
      2 b = np.array([1, 2, 3])
      3
----> 4 addition = a + b
      5 subtraction = a - b
      6 multiplication = a * b

ValueError: operands could not be broadcast together with shapes (4,) (3,)
```

Next steps: Explain error

```
import numpy as np

salary = np.array([30000, 40000, 50000, 60000])
increment = salary * 0.20
new_salary = salary + increment

print(salary)
print(increment)
```

```
print(new_salary)

bonus = new_salary * 0.05
print(bonus)
```

```
[30000 40000 50000 60000]
[ 6000.  8000. 10000. 12000.]
[36000. 48000. 60000. 72000.]
[1800. 2400. 3000. 3600.]
```

```
new_salary + bonus
```

```
array([37800., 50400., 63000., 75600.])
```

```
numbers = np.arange(2, 21, 2)
print(numbers)
```

```
[ 2  4  6  8 10 12 14 16 18 20]
```

```
import numpy as np

numbers = np.arange(2, 21, 2)
print(numbers)

result = numbers + 1
print(result)

result = result * 2
print(result)

print(len(numbers))
```

```
[ 2  4  6  8 10 12 14 16 18 20]
[ 3  5  7  9 11 13 15 17 19 21]
[ 6 10 14 18 22 26 30 34 38 42]
10
```

```
import numpy as np

a = np.arange(0, 11, 2)
b = np.linspace(0, 10, 6)

print(a)
print(b)

print(a * 2)
print(b * 2)
print(len(a))
print(len(b))
```

```
[ 0  2  4  6  8 10]
[ 0.  2.  4.  6.  8. 10.]
[ 0  4  8 12 16 20]
[ 0.  4.  8. 12. 16. 20.]
6
6
```

```
import numpy as np

a = np.arange(0, 11, 2)
b = np.linspace(0, 10, 6)

print(a)
print(b)

print(a * 2.0)
print(b * 2)
print(len(a))
print(len(b))
```

```
[ 0  2  4  6  8 10]
[ 0.  2.  4.  6.  8. 10.]
[ 0.  4.  8. 12. 16. 20.]
[ 0.  4.  8. 12. 16. 20.]
6
6
```

```
5/3
```

```
1.6666666666666667
```

```
5/5
```

```
1.0
```

HW

```
import numpy as np

a = np.array([1, 2, 3])
b = np.array([1.5, 2.5, 3.5])

print(a.dtype)
print(b.dtype)

c = a + b
print(c)
print(c.dtype)
```

```
import numpy as np

arr = np.array([10.5, 20.7, 30.9], dtype=int)

print(arr)
print(arr.dtype)

result = arr + 5
print(result)

print(result * 2)
```

```
import numpy as np

arr = np.array([0, 1, 2, 0, 5], dtype=bool)

print(arr)

result = arr * 1
print(result)

result = result + 5
print(result)

print(result.dtype)
```

```
import numpy as np

arr = np.zeros((3, 4))

print(arr)
print(arr + 10)
print(arr * 5)
print(arr + 100)
```